

MANIKANDAN SHANMUGAM

Aerospace Engineering Student (M2) – Propulsion & Energetics, ISAE-ENSMA

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Professional Profile

Second-year MSc student in Aeronautics and Space at ISAE-ENSMA, majoring in propulsion and energetics, with a background in mechanical engineering. Experience covers experimental research on two-phase heat transfer at the Institut Pprime, propulsion and CFD coursework projects, and four years of designing, building and flight-testing autonomous UAVs, including the Jury's Favourite Award at the Dassault UAV Challenge 2026. Seeking a six-month internship from March 2027 in propulsion, energetics, thermal engineering, CFD or autonomous aerial systems.

Education

MSc in Aeronautics and Space – Propulsion & Energetics (AME), M2 **2025 – 2027**
ISAE-ENSMA (École Nationale Supérieure de Mécanique et d'Aérotechnique) *Poitiers, France*

- Coursework: propulsion systems, combustion and thermochemistry, gas dynamics, heat transfer, fluid mechanics, numerical methods, instrumentation.

Bachelor of Engineering, Mechanical Engineering **2021 – 2025**
Loyola-ICAM College of Engineering and Technology *Chennai, India*

Research & Engineering Experience

Research Intern – Two-Phase Heat Transfer **Mar 2026 – Jul 2026**
Institut Pprime (CNRS, Université de Poitiers, ISAE-ENSMA) *Poitiers, France*

- Conducted experimental research on two-phase loop thermosyphons for passive thermal management.
- Improved the experimental setup for accuracy and repeatability, and performed high-speed flow visualisation with temperature, pressure and mass-flow measurements.
- Developed MATLAB routines to compute heat-transfer rates, thermal resistances and heat-transfer coefficients, applying Nusselt, Reynolds and Prandtl correlations to evaluate evaporator and condenser performance.
- Analysed the interaction between heat transfer and flow instabilities; co-authoring a manuscript based on the results, under internal review.

Propulsion & UAV Systems Engineer – Dassault UAV Challenge **Oct 2025 – Present**
EN SMAERO student team, ISAE-ENSMA *Poitiers, France*

- Won the Jury's Favourite Award (Prix Coup de cœur du jury) at the Dassault UAV Challenge 2026, among 28 teams, with a modular autonomous VTOL 4+1 UAV of 2 m wingspan.
- Carried out propulsion system design and CFD optimisation for the aircraft.
- Integrated on-board sensors with a SpeedyBee F405 flight controller and developed control algorithms in Python and C++, validated through iterative ground and flight testing.

R&D Engineering Intern – VTOL Design & Development **Jul 2024 – Mar 2025**
Team Reconnaissance, Hindustan Technology Business Incubator *Chennai, India*

- Designed and developed autonomous VTOL aerial systems on a Holybro flight controller paired with an NVIDIA Jetson Nano companion computer.
- Planned and executed flight-test campaigns with sensor integration and performance validation.
- Developed Python scripts and pipelines (NumPy, Pandas, Matplotlib) for flight-test data acquisition, processing and analysis.
- Integrated 3D-printed lightweight components and tuned mission planning and navigation for fixed-wing and multi-rotor platforms.

Automation & Systems Integration Intern **Feb 2024 – Apr 2024**
Loyola-ICAM College of Engineering and Technology *Chennai, India*

- Automated data-processing workflows in Python and developed interfaces connecting multiple data sources to centralised databases.

Wind Turbine Technician Intern **Jul 2023**
Litewind Ltd *Chennai, India*

- Assisted with wind-turbine installation, testing and maintenance, and studied electric motor and power-generation systems.

Coordinate Measuring Machine (CMM) Programmer Intern **Jun 2023 – Jul 2023**
Unique Measurement Service *Chennai, India*

- Programmed CMM measurement sequences from SolidWorks models for precision inspection of mechanical components, and prepared inspection reports.

Academic Projects – Propulsion, Energetics & CFD (ISAE-ENSMA)

ATREX Precooled Air-Breathing Engine – Cycle and Flight-Domain Analysis

Apr 2026

- Performed the complete thermodynamic cycle analysis of a liquid-hydrogen ATREX engine at 12,000 m and Mach 3 (intake, precooler, compressor map, combustion chamber, heat exchanger, turbine, nozzle).
- Computed thrust, specific impulse and efficiencies, and built the flight domain under thrust, drag and temperature limits, showing that a four-engine vehicle can reach 30 km and Mach 5–6.

Supersonic CFD – Diamond Airfoil, Busemann Biplane and Nozzle Flows (STAR-CCM+)

Apr 2026

- Simulated a double-wedge airfoil at Mach 4 with inviscid (Euler) and viscous turbulent (Spalart–Allmaras) models to separate wave drag from viscous drag, and studied the effect of thickness and trailing-edge geometry.
- Modelled a Busemann biplane at non-zero angle of attack and nozzle flows at pressure ratios of 8 and 12.

Transient Heat Conduction in a Sphere – Finite Volume Solver

Dec 2025

- Implemented an explicit finite-volume solver in spherical coordinates for a sphere with angle-dependent surface convection, derived its stability limit by von Neumann analysis and validated it with mesh-refinement tests.

H₂/O₂ Combustion Thermochemistry (Python, Cantera)

Nov 2025 – Jan 2026

- Modelled stoichiometric H₂/O₂ combustion for rocket-propulsion conditions: adiabatic flame temperature, chemical equilibrium and dissociation from 0.1 to 10⁴ MPa, cross-checked against an analytical estimate (3,498 K).

Automated Thermal Test Interface (MATLAB, Fortran)

Nov 2025 – Jan 2026

- Developed a GUI application for thermal testing with multi-sensor data acquisition, real-time visualisation and automated post-processing.

UAV & Autonomous Systems Projects

Founder & Team Captain – Team Phoenix (UAV Team)

Nov 2022 – Aug 2025

Team Turbonites LICET, Loyola-ICAM College of Engineering and Technology

Chennai, India

- Founded and led a 20-member UAV team, award winner at the SAE Autonomous Drone Development Challenge 2024 (payload category), managing project timeline, resources and technical design.
- Designed, built and flight-tested autonomous drones through a full cycle of CAD (SolidWorks), FEA, 3D printing and flight testing; integrated flight controller, IMU, GPS and sensors, with Gazebo flight simulation.
- Logged 100+ flight hours as drone pilot, including mission planning, drone mapping and aerial data collection.

Pixhawk MCP Server – Natural-Language Drone Control (Python, MAVLink, ArduPilot)

Ongoing

- Building a Python server that translates LLM tool calls into MAVLink commands (pymavlink) for a Pixhawk/ArduPilot drone: telemetry, arming, take-off, GPS waypoints, landing and return-to-launch, tested with ArduPilot SITL.

Custom Quadcopter Flight Controller with Automatic PID Tuning

Personal project

- Built a custom flight controller with automatic PID tuning of the control loops for plug-and-fly setup of quadcopters.

Technical Skills

Programming: Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB, C/C++, Fortran

Simulation & CFD: STAR-CCM+, ANSYS Fluent, Cantera, finite-volume methods, Gazebo, ArduPilot SITL

Thermal & Propulsion: Heat transfer, two-phase systems, thermodynamic cycle analysis, combustion thermochemistry, gas dynamics, experimental methods and instrumentation

CAD & Manufacturing: CATIA, SolidWorks, Fusion 360, FEA, 3D printing, CMM inspection

UAV & Embedded: Pixhawk (Cube Orange), Holybro, SpeedyBee F405, Radiolink CrossFlight, CC3D, ArduPilot, MAVLink, QGroundControl, Mission Planner, NVIDIA Jetson Nano, ESP32, ATmega328; UART, I2C, SPI, CAN, PWM, RS485

Tools: Git, Linux/UNIX, bash scripting

Awards & Certifications

- **Dassault UAV Challenge 2026** – Prix Coup de cœur du jury (Jury's Favourite Award), ENSMAERO team, among 28 teams.
- **SAE Autonomous Drone Development Challenge 2024** – Award winner (payload category), as founder and captain of Team Phoenix.
- **EU Certified Drone Pilot (DGAC)** – A1/A3 Open Category.

Leadership & Activities

- **Joint Secretary, ISHRAE Chennai Chapter** (Sep 2023 – Aug 2025) – Coordinated technical workshops and speaker sessions for a student organisation of 500+ members.
- **3D Printing Student Lead, LICET Fablab** (Jul 2023 – Aug 2025) – Organised fabrication and 3D-printing workshops and technical demonstrations.

Languages

English (Fluent) | French (A2) | Tamil (Native)